A LOWER BOUND FOR THE SIDON CONSTANT OF DIRICHLET POLYNOMIALS IN $\mathscr{H}^1$

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ABSTRACT. Let $S_1(x)$ be the supremum of $\sum_{n\le x}|a_n|$ over Dirichlet polynomials $\sum_{n\le x} a_n n^{-s}$ of norm at most 1 in the Hardy space $\mathscr{H}^1$. Helson's inequality gives $S_1(x) \le (1+o(1))\sqrt{x\log x}$, and a theorem of Harper gives $S_1(x) \gg \sqrt{x}(\log\log x)^{1/4}$. We prove that $S_1(x) \ge \sqrt{x}\exp(c(\log\log x)^{1/3})$ for an absolute constant $c>0$. The test polynomials have nonnegative multiplicative coefficients whose local exponents decrease from 3/2 to 1 along the primes, and the exponent 1/3 comes from the slowdown of branching random walks with decreasing variance.

1. INTRODUCTION

Let $\mathbb{T}^\infty$ be the countably infinite product of unit circles, equipped with its Haar measure, and let $p_1 < p_2 < \cdots$ be the primes. The Bohr lift of a Dirichlet polynomial $F(s) = \sum_{n\le x} a_n n^{-s}$ is the polynomial on $\mathbb{T}^\infty$ obtained by replacing p_j^{-s} by the jth coordinate z_j. For $1 \le p < \infty$, the Hardy space $\mathscr{H}^p$ of Dirichlet series is the closure of the Dirichlet polynomials in the norm

$$\|F\|_{\mathscr{H}^p} = \Big(\int_{\mathbb{T}^\infty} \Big|\sum_{n\le x} a_n z^{\kappa(n)}\Big|^p \,\mathrm{d}m_\infty(z)\Big)^{1/p},$$

where $n = \prod_j p_j^{\kappa_j(n)}$ and $z^{\kappa(n)} = \prod_j z_j^{\kappa_j(n)}$. For $x \ge 2$ we consider

$$S_1(x) = \sup\Big\{\sum_{n\le x}|a_n| \;:\; \Big\|\sum_{n\le x} a_n n^{-s}\Big\|_{\mathscr{H}^1} \le 1\Big\}. \tag{1}$$

The question is to determine the order of $S_1(x)$ as $x\to\infty$.

The corresponding quantity $S(x)$ for $\mathscr{H}^\infty$, with the supremum norm in place of the $\mathscr{H}^1$ norm, is the Sidon constant of Dirichlet polynomials of length x [1, (9.3)]. Its order is known precisely: $S(x) = \sqrt{x}\exp(-(1/\sqrt{2}+o(1))\sqrt{\log x\log\log x})$; see [1, Theorem 9.1], and [1, Section 9.3] for the history of this result. By the Cauchy–Schwarz inequality, the analogue of $S_1(x)$ for $\mathscr{H}^2$ equals $\sqrt{x}$. For $\mathscr{H}^1$ the trivial bounds are further apart. Helson's inequality [6]DGMS*Corollary 12.16 states that $(\sum_n |a_n|^2/d(n))^{1/2} \le \|F\|_{\mathscr{H}^1}$,

Date: October 13, 2026.

2020 *Mathematics Subject Classification.* Primary 30B50; Secondary 11K65, 30H10, 60G50.

Key words and phrases. Dirichlet polynomials, Hardy spaces, Sidon constant, random multiplicative functions, branching random walk.

where $d(n)$ is the number of divisors of n. Combined with the Cauchy–Schwarz inequality, it gives

$$S_1(x) \le \Big(\sum_{n\le x} d(n)\Big)^{1/2} = (1+o(1))\sqrt{x\log x}.$$

In the other direction, the Bohr lift identifies $\|F\|_{\mathscr{H}^1}$ with $\mathbb{E}|\sum_{n\le x} a_n f(n)|$, where f is a Steinhaus random multiplicative function (see Section 2). Harper [5, Theorem 1] proved that $\mathbb{E}|\sum_{n\le x} f(n)| \asymp \sqrt{x}(\log\log x)^{-1/4}$, and testing (1) on $\sum_{n\le x} n^{-s}$ therefore gives $S_1(x) \gg \sqrt{x}(\log\log x)^{1/4}$. A recent preprint of Hamdan [4, Theorem 1.1] bounds the first moment of $\sum_{n\le x} d_\alpha(n) f(n)$ for fixed $1<\alpha<2$, where d_α are the coefficients of $\zeta(s)^\alpha$; testing (1) on these polynomials gives $S_1(x) \gg_\varepsilon \sqrt{x}(\log\log x)^{3/2-\varepsilon}$, a consequence not stated there. The proof in [4] adopts a reduction from Gerspach's thesis [3, Proposition 3.6] and omits the details of its adaptation.

The main purpose of the present paper is to show that $S_1(x)/\sqrt{x}$ grows faster than every power of $\log\log x$.

Theorem 1. *There is an absolute constant $c>0$ with the following property. Suppose that x is sufficiently large. Then*

$$S_1(x) \ge \sqrt{x}\exp\big(c(\log\log x)^{1/3}\big).$$

To the best of our knowledge, Theorem 1 is new. It does not depend on [4] or [3]. The order of $S_1(x)$ remains open between Theorem 1 and $(1+o(1))\sqrt{x\log x}$.

Our approach. Let us briefly describe our approach. We test (1) on polynomials with nonnegative coefficients, for which the numerator is $\sum_{n\le x} a_n$ and the task is to bound the first moment from above. We take $a_n = g(n)$ multiplicative, with

$$\sum_{n=1}^{\infty} g(n)n^{-s} = \prod_p (1-p^{-s})^{-\alpha_p},$$

where α_p decreases from $3/2$ to 1 as $\log\log p$ runs from 0 to $t\approx\log\log x$, and $\alpha_p = 1$ for larger p. As a function of h, the logarithm of the Euler product $\prod_{p\le y}|1-f(p)p^{-1/2-ih}|^{-\alpha_p}$ behaves like a branching random walk in the time $j=\log\log p$, with variance $\alpha_j^2/2$ per unit of time. Fang and Zeitouni [2] and Maillard and Zeitouni [7] showed that when the variance decreases, the maximum of such a walk lags behind its first-order line by an amount of order $t^{1/3}$. We prove the first-moment half of this statement for our walk, as a penalised ballot estimate, and combine it with Harper's conditioning argument [5]. In contrast to the constant exponents in [5]Hamdan, the exponential tilt that makes the walk centred varies with j, and the variation produces a weight $\exp(2\sum_j \kappa_j B_j)$ with $\kappa_j \asymp 1/t$, which costs $\exp(-ct^{1/3})$ on walks that stay below a barrier. Since $\alpha_p = 1$ for large p, Harper's reduction applies without change.

Organization. The present paper is organized into six additional sections. In Section 2 we fix the coefficients g and estimate $\sum_{n\le x} g(n)$. In Section 3 we reduce the $\mathscr{H}^1$ norm to a short integral of an Euler product, following Harper. In Section 4 we introduce the walk and the change of measure. Section 5 contains the penalised ballot estimate, and Section 6 the crossing estimate and the proof of Theorem 1. Section 7 contains concluding remarks.

2. Preliminaries

Let $(f(p))_p$ be independent random variables, uniformly distributed on $\mathbb{T}$, and extend f completely multiplicatively to the positive integers. The law of $(f(p_j))_j$ is the Haar measure of $\mathbb{T}^\infty$, and under the Bohr lift n^{-s} corresponds to $f(n)$. Hence

$$\Big\|\sum_{n\le x} a_n n^{-s}\Big\|_{\mathscr{H}^1} = \mathbb{E}\Big|\sum_{n\le x} a_n f(n)\Big|. \tag{2}$$

Note that the $f(n)$ are orthonormal in L^2, and that for every real h the sequence $(f(p)p^{-ih})_p$ has the same law as $(f(p))_p$.

Throughout, C and c denote positive absolute constants, which may change from line to line. Let $T = \log\log x$, let $t = \lfloor T - 2\log T\rfloor$ and $y = \exp(e^t)$. Then

$$\frac{\log x}{eT^2} \le \log y = e^t \le \frac{\log x}{T^2}. \tag{3}$$

Let I_1 be the set of primes $p \le e^e$, and for $2 \le j \le t$ let I_j be the set of primes in $(\exp(e^{j-1}), \exp(e^j)]$. By Mertens' theorem [10, §I.1.6],

$$\sum_{p\le z} \frac{1}{p} = \log\log z + M + O\Big(\frac{1}{\log z}\Big), \tag{4}$$

where M is the Meissel–Mertens constant. Hence there is an absolute j_0 such that $1/2 \le \sum_{p\in I_j} 1/p \le 2$ for $j \ge j_0$, and $\sum_{p\in I_j} 1/p \le 2$ for every j. We put

$$\alpha_j = \frac{3}{2} - \frac{j-1}{2(t-1)}\ (1\le j\le t), \qquad \alpha_p = \alpha_j\ (p\in I_j), \qquad \alpha_p = 1\ (p>y).$$

So α_j decreases linearly from $3/2$ to 1, and $1 \le \alpha_p \le 3/2$. Let g be the multiplicative function with $g(p^k) = \binom{\alpha_p+k-1}{k}$. Then $\sum_n g(n)n^{-s} = \prod_p(1-p^{-s})^{-\alpha_p}$, $g \ge 0$, $g(n) = 1$ when every prime factor of n exceeds y, and $g(p^k) \le k+1$.

We first record that the coefficient sum of g is as large as the Euler product predicts.

Lemma 1. *Suppose that x is sufficiently large. Then*

$$\sum_{n\le x} g(n) \ge \frac{x}{4}\exp\Big(\sum_{p\le y}\frac{\alpha_p - 1}{p}\Big).$$

Proof. Write $g = 1 * h$, where h is the multiplicative function with $\sum_n h(n)n^{-s} = \prod_{p\le y}(1-p^{-s})^{-(\alpha_p-1)}$. Then $h \ge 0$, and h is supported on integers whose prime factors are at most y. Since $\lfloor x/m\rfloor \ge x/(2m)$ for $m \le x$, we get

$$\sum_{n\le x} g(n) = \sum_{m\le x} h(m)\Big\lfloor \frac{x}{m}\Big\rfloor \ge \frac{x}{2}\Big(P - \sum_{m>x}\frac{h(m)}{m}\Big), \qquad P = \prod_{p\le y}\Big(1-\frac{1}{p}\Big)^{-(\alpha_p-1)}.$$

Let $\delta = 1/(2\log y)$, so that $p^\delta \le e^{1/2}$ for $p \le y$. By Rankin's trick,

$$\sum_{m>x}\frac{h(m)}{m} \le x^{-\delta}\sum_m \frac{h(m)}{m^{1-\delta}} = x^{-\delta}\prod_{p\le y}(1-p^{\delta-1})^{-(\alpha_p-1)} \le CPx^{-\delta}.$$

The last step holds because $p^\delta - 1 \le C\delta\log p$ and $\delta\sum_{p\le y}(\log p)/p \le C$. By (3), $x^{-\delta} \le e^{-T^2/2}$. Finally, $P \ge \exp(\sum_{p\le y}(\alpha_p-1)/p)$, since $-\log(1-1/p) \ge 1/p$. □

3. Harper's conditioning argument

In view of (2) and Lemma 1, Theorem 1 reduces to an upper bound for the first moment of $\sum_{n\le x} g(n)f(n)$. Let

$$F_y(s) = \prod_{p\le y}(1-f(p)p^{-s})^{-\alpha_p}, \qquad \mathcal{Z} = \int_0^1 |F_y(1/2+ih)|^2\,\mathrm{d}h.$$

The next result reduces the first moment to $\mathbb{E}\mathcal{Z}^{1/2}$. It is Harper's conditioning argument [5], in the form given by Hamdan [4, Section 2.1]. We include the proof, since the coefficients g are not covered by these papers. The point is that $g = 1$ on integers without prime factors below y.

Lemma 2. *Suppose that x is sufficiently large. Then*

$$\mathbb{E}\Big|\sum_{n\le x} g(n)f(n)\Big| \le C\sqrt{x}(e^{-T^2/16}(\log x)^5 + te^{-t/2}\,\mathbb{E}\mathcal{Z}^{1/2} + e^{-3t/2}).$$

Proof. Call n y-smooth if all its prime factors are at most y, and y-rough if all its prime factors exceed y. Write each n uniquely as $n = n_1n_2$ with n_1 y-rough and n_2 y-smooth. Then $g(n) = g(n_2)$. Let $S(u) = \sum_{n_2\le u} g(n_2)f(n_2)$, the sum running over y-smooth n_2. Then

$$X = \sum_{n\le x} g(n)f(n) = \sum_{n_1\le x} f(n_1)S(x/n_1),$$

where n_1 runs over y-rough integers, here and below. Let $\mathcal{F}_y$ be the σ-algebra generated by $(f(p))_{p\le y}$. The values $f(n_1)$ are independent of $\mathcal{F}_y$ and orthonormal. Hence, by the conditional Jensen inequality and the

subadditivity of the square root,

$$\mathbb{E}|X| \le \mathbb{E}\big(\mathbb{E}[|X|^2 \mid \mathcal{F}_y]\big)^{1/2} = \mathbb{E}\Big(\sum_{n_1\le x} |S(x/n_1)|^2\Big)^{1/2}$$
$$\le \mathbb{E}\Big(\sum_{n_1\le x^{3/4}} |S(x/n_1)|^2\Big)^{1/2} + \mathbb{E}\Big(\sum_{x^{3/4}<n_1\le x} |S(x/n_1)|^2\Big)^{1/2}.$$

We first treat small n_1. Let $u \ge x^{1/4}$ and $\delta = 1/(2\log y)$. Then $u^{-\delta} \le e^{-T^2/8}$ by (3). By orthonormality and Rankin's trick,

$$\mathbb{E}|S(u)|^2 = \sum_{n_2\le u} g(n_2)^2 \le u^{1-\delta} \prod_{p\le y} \sum_{k\ge 0} (k+1)^2 p^{-k(1-\delta)} \le C u\, e^{-T^2/8} (\log y)^8.$$

Here we used $\sum_k (k+1)^2 w^k = (1+w)/(1-w)^3 \le e^{8w}$ for $0 \le w \le e^{1/2}/2$, and $\sum_{p\le y} p^{\delta-1} \le \log\log y + C$. Summing over $n_1 \le x^{3/4}$ and applying Jensen's inequality, we find that the first term is at most $C\sqrt{x}(\log x)^{1/2} e^{-T^2/16} (\log y)^4$.

We turn to large n_1. Put $r = \lfloor x/n_1 \rfloor$, so that $1 \le r < x^{1/4}$ and $S(x/n_1) = S(r)$. The interval $(x/(r+1), x/r]$ has length at least $x^{1/2}/2 \ge y^3$ by (3). Selberg's upper bound sieve [8, §3.2] shows that it contains at most $Cx/(r^2 \log y)$ integers that are y-rough. Since S is constant on $[r, r+1)$, we get

$$\sum_{x^{3/4}<n_1\le x} |S(x/n_1)|^2 \le \frac{Cx}{\log y} \sum_{r\ge 1} \frac{|S(r)|^2}{r^2} \le \frac{4Cx}{\log y} \int_1^\infty \frac{|S(u)|^2}{u^2}\,\mathrm{d}u.$$

Since $\sum_{n_2} g(n_2) n_2^{-1/4} < \infty$, we have $|S(u)| \le C u^{1/4}$, where C may depend on x. Hence the function $v \mapsto S(e^v) e^{-v/2}$ lies in $L^2(0,\infty)$. By Fubini's theorem its Fourier transform is $\int_1^\infty S(u) u^{-3/2-i\tau}\,\mathrm{d}u = F_y(1/2+i\tau)/(1/2+i\tau)$, and Plancherel's theorem gives

$$\int_1^\infty \frac{|S(u)|^2}{u^2}\,\mathrm{d}u = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|F_y(1/2+i\tau)|^2}{|1/2+i\tau|^2}\,\mathrm{d}\tau.$$

Let $H = (\log y)^{10}$. For $\tau \in [n, n+1]$ we have $|1/2 + i\tau| \ge (|n|+1)/4$. Since $(f(p)p^{-in})_p$ has the law of $(f(p))_p$, the integral $\int_n^{n+1} |F_y(1/2+i\tau)|^2\,\mathrm{d}\tau$ has the law of $\mathcal{Z}$, and $\mathbb{E}|F_y(1/2+i\tau)|^2 = \mathbb{E}|F_y(1/2)|^2$ for every τ. Hence, by subadditivity of the square root and Jensen's inequality,

$$\mathbb{E}\Big(\int_{\mathbb{R}} \frac{|F_y(1/2+i\tau)|^2}{|1/2+i\tau|^2}\,\mathrm{d}\tau\Big)^{1/2} \le \sum_{|n|\le H} \frac{4\,\mathbb{E}\mathcal{Z}^{1/2}}{|n|+1} + \Big(\frac{2\,\mathbb{E}|F_y(1/2)|^2}{H}\Big)^{1/2}$$
$$\le Ct\,\mathbb{E}\mathcal{Z}^{1/2} + \frac{C}{\log y}.$$

Here we used $\log H \le Ct$ and $\mathbb{E}|F_y(1/2)|^2 = \prod_{p\le y} \sum_k g(p^k)^2 p^{-k} \le C(\log y)^8$. Since $\log y = e^t$, the second term is at most $C\sqrt{x}(te^{-t/2}\mathbb{E}\mathcal{Z}^{1/2} + e^{-3t/2})$. Finally, $(\log x)^{1/2}(\log y)^4 \le (\log x)^5$. □

4. The Euler product walk

In this section we write $\log|F_y(1/2+ih)|$ as a random walk and find an exponential tilt under which the walk is centred. For $h \in \mathbb{R}$ and $1 \le j \le t$ put

$$\ell_p(h) = -\log|1-f(p)p^{-1/2-ih}|, \quad Y_j(h) = \alpha_j \sum_{p\in I_j} \ell_p(h), \quad W_j(h) = \sum_{i\le j} Y_i(h),$$

so that $\mathcal{Z} = \int_0^1 e^{2W_t(h)}\,\mathrm{d}h$. For each h, the process $(W_j(h))_{j\le t}$ has the law of $(W_j(0))_{j\le t}$, and $Y_1(0),\dots,Y_t(0)$ are independent. We write $\ell_p = \ell_p(0)$, $Y_j = Y_j(0)$ and $W_j = W_j(0)$. Since $p^{-1/2} \le 2^{-1/2}$, we have $|\ell_p| \le -\log(1-p^{-1/2}) \le 4p^{-1/2}$.

The tilt is exact. Expanding $|1-f(p)p^{-1/2}|^{-\beta}$ in powers of $f(p)$ gives $\mathbb{E}\,e^{\beta\ell_p} = \sum_k \binom{\beta/2+k-1}{k}^2 p^{-k}$. Taking $\beta = 2$, and differentiating with respect to β at $\beta = 2$, we find that

$$\mathbb{E}\,e^{2\ell_p} = \Big(1-\frac{1}{p}\Big)^{-1}, \qquad \mathbb{E}\big[\ell_p e^{2\ell_p}\big] = \sum_{k\ge1}\frac{H_k}{p^k}, \tag{5}$$

where $H_k = \sum_{i\le k} 1/i$. Let $\theta_j = 2/\alpha_j$, so that $\theta_j Y_j = 2\sum_{p\in I_j}\ell_p$, and define a probability measure $\mathbb{Q}$ by

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \exp\Big(\sum_{j\le t}\theta_j Y_j - \Lambda\Big), \qquad \Lambda = -\sum_{p\le y}\log\Big(1-\frac{1}{p}\Big).$$

Under $\mathbb{Q}$ the $f(p)$, $p\le y$, are independent, and $f(p)$ has density $(1-1/p)|1-zp^{-1/2}|^{-2} \ge 1/6$ with respect to the Haar measure of $\mathbb{T}$. Put

$$\mu_j = \mathbb{E}_{\mathbb{Q}}Y_j, \qquad \varphi(j) = \sum_{i\le j}\mu_i, \qquad \xi_j = Y_j - \mu_j, \qquad B_j = \sum_{i\le j}\xi_i = W_j - \varphi(j).$$

By (5), $\mathbb{E}_{\mathbb{Q}}\ell_p = (1-1/p)\sum_k H_k p^{-k}$ lies between $(1-1/p)/p$ and $1/(p-1)$. Together with (4), this gives

$$\Big|\varphi(t) - \sum_{p\le y}\frac{\alpha_p}{p}\Big| \le C, \qquad \Big|\Lambda_J - \sum_{i\le J}\theta_i\mu_i + J\Big| \le C \quad (1\le J\le t), \tag{6}$$

where $\Lambda_J = -\sum_{p\in I_1\cup\dots\cup I_J}\log(1-1/p)$. For the second bound, note that $\sum_{i\le J}\theta_i\mu_i = 2\sum_{p\in I_1\cup\dots\cup I_J}\mathbb{E}_{\mathbb{Q}}\ell_p$. Finally, let $\kappa_j = 1/\alpha_{j+1} - 1/\alpha_j$ for $1\le j<t$. Then

$$\kappa_j \ge \frac{2}{9t}, \qquad \sum_{i<j}\kappa_i = \frac{1}{\alpha_j} - \frac{1}{\alpha_1}. \tag{7}$$

The next lemma transfers events for the walk under $\mathbb{P}$ to the centred walk under $\mathbb{Q}$. The varying tilt appears as the weight $\exp(2\sum_j \kappa_j B_j)$.

Lemma 3. *Suppose that $1\le J\le t$, and that $\Phi \ge 0$ is a function of $W_1,\dots,W_J$. Then*

$$\mathbb{E}\,\Phi \le Ce^{-J}\,\mathbb{E}_{\mathbb{Q}}\Big[\Phi\exp\Big(-\theta_J B_J + 2\sum_{j<J}\kappa_j B_j\Big)\Big].$$

Proof. The factors of $\mathrm{d}\mathbb{P}/\,\mathrm{d}\mathbb{Q}$ with $j > J$ have $\mathbb{Q}$-expectation 1 and are independent of Φ. Hence $\mathbb{E}\Phi = \mathbb{E}_{\mathbb{Q}}[\Phi \exp(\Lambda_J - \sum_{j\le J} \theta_j Y_j)]$. Summation by parts gives

$$\sum_{j\le J} \theta_j Y_j = \sum_{j\le J} \theta_j \mu_j + \theta_J B_J - 2\sum_{j<J} \kappa_j B_j,$$

since $\theta_{j+1} - \theta_j = 2\kappa_j$. The second bound in (6) finishes the proof. □

We also need two standard properties of the increments under $\mathbb{Q}$: a sub-Gaussian bound, and a lower bound for the probability of a large rise over a long stretch.

Lemma 4. *The ξ_j are independent and centred under $\mathbb{Q}$. Moreover:*

1. *There is an absolute constant C_1 such that $\mathbb{E}_{\mathbb{Q}} e^{s\xi_j} \le e^{C_1 s^2}$ for every real s and every j.*
2. *For every $M > 0$ there are $p_M > 0$ and L_M such that $\mathbb{Q}(B_{k+L} - B_k \ge M\sqrt{L}) \ge p_M$ whenever $L \ge L_M$ and $k \ge j_0$.*

Proof. (1) Each $\alpha_j \ell_p$ takes values in an interval of length $16p^{-1/2}$, and $\sum_{p\in I_j} 1/p \le 2$. Hence the bound follows from Hoeffding's lemma [11, Example 2.5.8(c), Propositions 2.5.2 and 2.6.1].

(2) The increment $B_{k+L} - B_k$ is a sum over primes of independent centred terms bounded by $16p^{-1/2}$, so the sum of their third absolute moments is at most C. Since the $\mathbb{Q}$-density of $f(p)$ is at least $1/6$ and $\alpha_j \ge 1$, we have

$$\mathrm{Var}_{\mathbb{Q}}(\alpha_j \ell_p) \ge \frac{\mathrm{Var}_{\mathbb{P}}(\ell_p)}{6} = \sum_{k\ge 1} \frac{p^{-k}}{12k^2} \ge \frac{1}{12p}.$$

Hence the variance σ^2 of $B_{k+L} - B_k$ satisfies $L/24 \le \sigma^2 \le CL$. By the Berry–Esseen inequality for independent summands [9, Chapter V, §2],

$$\mathbb{Q}(B_{k+L} - B_k \ge M\sqrt{L}) \ge 1 - \Phi_{\mathcal{N}}(5M) - CL^{-3/2},$$

where $\Phi_{\mathcal{N}}$ is the standard normal distribution function. □

5. A penalised ballot estimate

The following estimate is the source of the factor $\exp(-ct^{1/3})$. A walk that stays below a barrier pays the weight $\exp(2\sum_j \kappa_j B_j)$ when it stays far below, and has a fixed probability of crossing when it stays close. We split time into blocks of length $t^{2/3}$, on which these two costs balance.

Lemma 5. *Suppose that $A \ge 0$. Then*

$$\mathbb{E}_{\mathbb{Q}}\Big[\exp\Big(2\sum_{j<t} \kappa_j (B_j - A)\Big);\, B_j \le A \text{ for every } j \le t\Big] \le Ce^{-ct^{1/3}}.$$

Proof. Let $V_j = B_j - A$, so that $V_0 = -A \le 0$. On the event in question every factor $e^{2\kappa_j V_j}$ is at most 1. Let $L = \lceil t^{2/3} \rceil$, $s_m = j_0 + mL$ and $N = \lfloor (t - 1 - j_0)/L \rfloor$, so that $N \ge t^{1/3}/2$ for large t. We drop the factors and the constraints outside the blocks $(s_m, s_m + L]$, $0 \le m < N$, except

$V_{s_0} \le 0$. Conditioning successively on V_{s_m} and using the Markov property, we find that the expectation is at most ρ^N, where

$$\rho = \sup_{m<N} \sup_{v\le 0} \mathbb{E}_{\mathbb{Q}}\Big[\exp\Big(2 \sum_{s_m<j\le s_m+L} \kappa_j V_j\Big); \\ V_j \le 0 \ (s_m < j \le s_m + L) \ \Big| \ V_{s_m} = v\Big].$$

Fix M with $e^{-M^2/(16C_1)} + e^{-2M/9} \le 1/2$, with C_1 as in Lemma 4. Let $m < N$ and $v \le 0$.

Suppose first that $v > -M\sqrt{L}$. On the event, the walk does not rise by $M\sqrt{L}$ over the block. By part (2) of Lemma 4, the quantity is at most $1 - p_M$ once $L \ge L_M$.

Suppose next that $v \le -M\sqrt{L}$. Then either $\max_{i\le L}(V_{s_m+i} - v) \ge M\sqrt{L}/2$, or $V_j \le -M\sqrt{L}/2$ throughout the block. By Doob's maximal inequality applied to $e^{\lambda(V_{s_m+i}-v)}$ with $\lambda = M/(4C_1\sqrt{L})$, and part (1) of Lemma 4, the first event has probability at most $e^{-M^2/(16C_1)}$. On the second event, by (7), the weight is at most $\exp(-2L \cdot (2/(9t)) \cdot M\sqrt{L}/2) \le e^{-2M/9}$, since $L^{3/2} \ge t$.

Hence $\rho \le \max(1 - p_M, 1/2) < 1$ for large t, and $\rho^N \le e^{-ct^{1/3}}$. □

For $A \ge 0$ let

$$G_A = \{h \in [0,1] : W_j(h) \le \varphi(j) + A \ (j \le t)\}, \qquad \mathcal{Z}(A) = \int_{G_A} e^{2W_t(h)} \, \mathrm{d}h.$$

Armed with Lemmas 3 and 5, we can bound the first moment of the part of $\mathcal{Z}$ that comes from points below the barrier.

Lemma 6. *Suppose that $A \ge 0$. Then*

$$\mathbb{E}\mathcal{Z}(A) \le C \exp\Big(2\varphi(t) - t + 2A\Big(1 - \frac{1}{\alpha_1}\Big) - ct^{1/3}\Big).$$

Proof. By Fubini's theorem and stationarity in h, $\mathbb{E}\mathcal{Z}(A) = \mathbb{E}[e^{2W_t}; W_j \le \varphi(j) + A \text{ for every } j \le t]$. We apply Lemma 3 with $J = t$ and $e^{2W_t} = e^{2\varphi(t)+2B_t}$. Since $2-\theta_t = 2(1-1/\alpha_t)$, the exponent becomes, with $V_j = B_j - A$ and by (7),

$$2\Big(1 - \frac{1}{\alpha_t}\Big)(A + V_t) + 2\sum_{j<t} \kappa_j (A + V_j) \le 2A\Big(1 - \frac{1}{\alpha_1}\Big) + 2\sum_{j<t} \kappa_j V_j,$$

because $\alpha_t \ge 1$ and $V_t \le 0$ on the event. Now Lemma 5 finishes the proof. □

6. Crossings and the proof of Theorem 1

It remains to control the probability that some $h \in [0,1]$ crosses the barrier. Let $\mathcal{B}_A$ be the event that $G_A \ne [0,1]$. We bound the probability of a crossing at a single h by Lemma 3, and pass to all $h \in [0,1]$ with a bound for the derivative of W_j in h, which costs a factor $t^{3/2}$.

Lemma 7. *Suppose that $A \ge 0$. Then $\mathbb{P}(\mathcal{B}_A) \le Ct^{3/2}(1 + \sqrt{A})e^{-2A/\alpha_1}$.*

Proof. Let $D_j = e^{-j}\sup_{h\in[0,1]}|\partial_h W_j(h)|$. We first show that there is an absolute constant $C_2 \geq 1$ such that

$$\mathbb{P}\big(D_j > C_2(\sqrt{j}+u)\big) \leq 4e^{-u^2} \qquad (u \geq 0). \tag{8}$$

Note that $\partial_h \ell_p(h)$ is a centred function of $f(p)$, since $\mathbb{E}\ell_p(h) = 0$ for every h. Moreover, $|\partial_h \ell_p(h) - \partial_h \ell_p(h')| \leq C(\log p)p^{-1/2}\min(1, |h-h'|\log p)$. By Hoeffding's inequality [11, Example 2.5.8(c) and Proposition 2.6.1] and the bound $\sum_{p\leq z}(\log p)^m/p \leq C_m(\log z)^m$, the process $e^{-j}\partial_h W_j(h)$ has sub-Gaussian increments with respect to the metric $d(h,h') = C\min(1, e^j|h-h'|)$, and its value at 0 is sub-Gaussian with bounded norm. The covering numbers of $[0,1]$ satisfy $N([0,1],d,\varepsilon) \leq 1 + Ce^j/\varepsilon$, so Dudley's integral is at most $C\sqrt{j} + C$, and the tail bound in [11, Theorem 8.1.6] gives (8).

On $\mathcal{B}_A$, let J be the least $j \leq t$ for which some $h \in [0,1]$ has $W_j(h) > \varphi(j) + A$. Let $k \geq 1$, and let E_j be the set of $h \in [0,1]$ such that $W_j(h) > \varphi(j)+A-1$ and $W_i(h) \leq \varphi(i) + A$ for every $i < j$. On the event $\{J = j,\ D_j \leq k\}$, pick $h_0 \in [0,1]$ with $W_j(h_0) > \varphi(j) + A$. Every $h \in [0,1]$ with $|h - h_0| \leq e^{-j}/k$ lies in E_j, so E_j has measure at least e^{-j}/k. By Markov's inequality, Fubini's theorem, stationarity in h and Lemma 3,

$$\begin{aligned}\mathbb{P}(J=j) &\leq \mathbb{P}(D_j > k) + ke^j\,\mathbb{P}(0 \in E_j)\\ &\leq \mathbb{P}(D_j > k)\\ &\quad + Ck\,\mathbb{E}_{\mathbb{Q}}\Big[e^{-\theta_j B_j + 2\sum_{i<j}\kappa_i B_i};\ B_i \leq A\ (i<j),\ B_j > A-1\Big].\end{aligned}$$

On this event, by (7), the exponent is at most $-\theta_j(A-1) + 2A(1/\alpha_j - 1/\alpha_1) \leq 2 - 2A/\alpha_1$. We take $u^2 = 2A + 2\log t$ and $k = C_2(\sqrt{t}+u)$ in (8). Then

$$\mathbb{P}(J=j) \leq \frac{4e^{-2A}}{t^2} + C(\sqrt{t}+\sqrt{A})e^{-2A/\alpha_1}.$$

Summing over $j \leq t$ and using $\alpha_1 > 1$, we obtain the claim. □

We are now ready to prove Theorem 1.

Proof of Theorem 1. The field W is bounded, so $\mathcal{B}_k$ is empty for large k. On the complement of $\mathcal{B}_0$ we have $\mathcal{Z} = \mathcal{Z}(0)$, and on $\mathcal{B}_k \setminus \mathcal{B}_{k+1}$ we have $\mathcal{Z} = \mathcal{Z}(k+1)$. Hence, by the Cauchy–Schwarz inequality and Lemmas 6 and 7,

$$\begin{aligned}\mathbb{E}\mathcal{Z}^{1/2} &\leq (\mathbb{E}\mathcal{Z}(0))^{1/2} + \sum_{k\geq 0}\mathbb{P}(\mathcal{B}_k)^{1/2}(\mathbb{E}\mathcal{Z}(k+1))^{1/2}\\ &\leq Ct^{3/4}e^{\varphi(t)-t/2-ct^{1/3}}\sum_{k\geq 0}(1+k)^{1/4}e^{-k(2/\alpha_1-1)}.\end{aligned}$$

The series converges because $\alpha_1 = 3/2 < 2$. By (6), Lemma 2 gives

$$\begin{aligned}\mathbb{E}\Big|\sum_{n\leq x} g(n)f(n)\Big| &\leq C\sqrt{x}\,t^{7/4}\exp\Big(\sum_{p\leq y}\frac{\alpha_p}{p} - t - ct^{1/3}\Big)\\ &\quad + C\sqrt{x}(e^{-T^2/16}(\log x)^5 + e^{-3t/2}).\end{aligned}$$

By (4), $\sum_{p\le y} 1/p = t + O(1)$. Hence, by Lemma 1, the ratio of $\sum_{n\le x} g(n)$ to each of the three terms on the right-hand side is at least $c\sqrt{x}\,t^{-7/4}e^{ct^{1/3}}$, $\sqrt{x}\,e^{T^2/32}$ and $c\sqrt{x}\,e^{3t/2}$, respectively. Since $g \ge 0$, the definition (1) and (2) give $S_1(x) \ge c\sqrt{x}\,t^{-7/4}e^{ct^{1/3}}$. Since $t \ge T - 2\log T - 1$, this proves Theorem 1. □

7. Concluding remarks

The size of the constant. The constant c in Theorem 1 is not explicit. The exponent $1/3$ is the slowdown found by Fang and Zeitouni [2] and computed precisely by Maillard and Zeitouni [7] for branching Brownian motion with decreasing variance. If their asymptotics transferred to the walk W, the best profile would make $\alpha_j^{3/2}$ linear, decreasing from 2 to 1, and give c close to $(\mathrm{a}_1/2)((2/3)(2^{3/2}-1))^{2/3} \approx 1.33$, where $-\mathrm{a}_1$ is the largest zero of the Airy function. This is a heuristic. Our argument requires $\alpha_1 < 2$, since the series in the proof of Theorem 1 has ratio $e^{-(2/\alpha_1 - 1)}$; a profile starting at 2 would require estimates that are uniform as $\alpha_1 \to 2$.

The range $2 \le p < \infty$. Let $S_p(x)$ be defined by (1) with $\mathscr{H}^p$ in place of $\mathscr{H}^1$. For $2 \le p < \infty$ we have $S_p(x) \asymp_p \sqrt{x}$. Indeed, $\sum_{n\le x}|a_n| \le \sqrt{x}\|F\|_{\mathscr{H}^2} \le \sqrt{x}\|F\|_{\mathscr{H}^p}$. Conversely, for independent random signs ε_n, Khintchine's inequality applied for each fixed $z \in \mathbb{T}^\infty$ and Fubini's theorem give $\mathbb{E}_\varepsilon\|\sum_{n\le x}\varepsilon_n n^{-s}\|_{\mathscr{H}^p}^p \le C_p x^{p/2}$, so some choice of signs gives $\sum_{n\le x}|a_n| = x$ and $\|F\|_{\mathscr{H}^p} \le C_p\sqrt{x}$. The range $1 \le p < 2$ is therefore the interesting one, and Theorem 1 shows that $S_1(x)/\sqrt{x}$ is unbounded, in contrast with $S(x)/\sqrt{x} \to 0$ for $\mathscr{H}^\infty$.

The order of $S_1(x)$. We do not expect nonnegative multiplicative coefficients with local factors $(1-p^{-s})^{-\alpha_p}$ to give more than $\sqrt{x}\exp(O((\log\log x)^{1/3}))$. On the other side, an improvement of the upper bound requires cancellation beyond Helson's inequality.

Problem 1. *Determine the order of $S_1(x)$. In particular, is $S_1(x) = o(\sqrt{x\log x})$?*

Acknowledgements. AI assistance was used in developing and checking the argument and in preparing the exposition; the author has verified the proof.

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